



Tire Ridging – Reaction to Longitudinal Road Ridges

1 Scope

Note: Nothing in this standard supercedes applicable laws and regulations.

Note: In the event of conflict between the English and domestic language, the English language shall take precedence.

1.1 Purpose. Determine the vehicle's reaction to driving, at a tangent, over longitudinal ridges such as track grooves, pavement joints, etc.

1.2 Foreword. Not applicable.

1.3 Applicability. For all tires with an aspect ratio of < 65% (height/width).

2 References

Note: Only the latest approved standards are applicable unless otherwise specified.

2.1 External Standards/Specifications.

None

2.2 GM Standards/Specifications.

GMW14264 GMW16590

2.3 Additional References.

Michelin/Ladoux Test Track, France

Pferdsfeld Test Track, Germany

3 Resources

3.1 Facilities. Suitable track with track grooves or road irregularities, joints. Examples of such surfaces exist on Test Track 17 at Michelin/Ladoux or at Pferdsfeld.

3.2 Equipment.

3.2.1 Axle measuring stand.

3.2.2 Tire pressure gage.

3.3 Test Vehicle/Test Piece. Vehicle with standard chassis and specified geometry for the test tires.

3.3.1 Test tires and corresponding reference tires.

3.4 Test Time. Not applicable.

3.5 Test Required Information. Not applicable.

3.6 Personnel/Skills. Not applicable.

4 Procedure

4.1 Preparation.

4.1.1 The test tire shall be mounted on the approved rims.

4.1.2 The test tire shall be mounted on the vehicle and the tire adjusted to partial load pressure.

4.2 Conditions.

4.2.1 Environmental Conditions. According to Section 4.3.

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4.2.2 Test Conditions. Deviations from the requirements of this standard shall have been agreed upon. Such requirements shall be specified on component drawings, test certificates, reports, etc.

4.3 Instructions.

4.3.1 The vehicle shall be driven at various speeds up to V_{maximum} on the relevant test tracks, track grooves, joints, etc., which shall be crossed several times at a tangent ± 5 degrees to ± 10 degrees.

4.3.2 The reaction of the vehicle when crossing the track grooves, etc., shall be evaluated subjectively according to GMW14264 and recorded on Data Sheet A1 in Appendix A.

5 Data

5.1 Calculations. Not applicable.

5.2 Interpretation of Results. The vehicle shall cross the track grooves without abrupt sideways movements. If the car jerks sideways when crossing the grooves, making steering corrections necessary, these reactions shall be evaluated by means of comparative drives with narrower reference tires according to GMW14264.

5.3 Test Documentation. The data/information to input on the Sample Report B1 in Appendix B is GM Proprietary. The test results documentation should include:

- Test results report (generated by Global Tire Wheel Test System (GTWTS) if submitting Extensible Markup Language (XML) files)^{Note 1}
- Data Sheet A1 shown in Appendix A

Note 1: For data being loaded into the GTWTS, submit electronically the test results data files (XML format) and other supporting files listed. GMW16590 defines the electronic file details (e.g., file types, file naming convention, and detailed file formats) for the GTWTS.

6 Safety

This standard may involve hazardous materials, operations, and equipment. This standard does not propose to address all the safety problems associated with its use. It is the responsibility of the user of this standard to establish appropriate safety and health practices and determine the applicability of regulatory limitations prior to use.

7 Notes

7.1 Glossary. Not applicable.

7.2 Acronyms, Abbreviations, and Symbols.

GSSLT	Global Subsystem Leadership Team
GTWTS	Global Tire Wheel Test System
V	Velocity
XML	Extensible Markup Language

8 Coding System

This standard shall be referenced in other documents, drawings, etc., as follows:

Test to GMW15002

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9 Release and Revisions

This standard was originated in January 2006. It was first approved by the Global Tire and Wheel Leadership team in September 2006. It was first published in May 2007.

Issue	Publication Date	Description (Organization)
1	MAY 2007	Initial publication.
2	JAN 2013	Revise to current template. (Tire and Wheel GSSLT)
3	FEB 2013	Reversion for Template Exception (3.1.1 and 3.1.2 removed). (Tire and Wheel GSSLT)

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Data Sheet A1: Tire Ridging Test Data

**Tire-Wheel
Systems**

PQMS Activity Number:

[illegible]

GM Proprietary

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Date Printed:

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Appendix B

Table B1: Sample GM Global Database Summary Report

GMW15002 Tire Ridging Report



Test Request Number:
PQMS Activity Number:

TEST REQUEST DETAILS		
Model Year / Program:	Requester:	
Test Purpose:	Requester's Region:	
TIRE PART DETAILS		
Supplier / Trade Name:	Design Vehicle:	Construction Number:
Load Index / Load (N):	Tire Type:	Tire Size/Load Range:
Speed Rating:	Part Status Code:	Sidewall:
Part Numbers:	Tire Condition:	
TEST CONDITIONS		
Gross Axle Weight Rating Front (kg):	Requested Front Pressure (kPa):	
Gross Axle Weight Rating Rear (kg):	Requested Rear Pressure (kPa):	
TR Comment:	Requested Rim Width (in):	
TEST RESULTS		
Tester / Test Site:	Number of Tires:	
Test Date:	Machine / ID:	
TIP Approved:	Technician:	
Comment:		
Published by	Date:	
FILE ATTACHMENTS		

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